Chapter 3 — The Persistence Core & Single Source of Truth

> **Part I — Core Infrastructure** · Complexity: **Complex** (≥40 interview questions)
>
> **Prerequisites:** Ch 1 (Networking Core — `Outcome` / `AppError`, DTOs, `safeApiCall`), Ch 2 (reliability, optional). This chapter **defines** the persistence layer and the **Single Source of Truth (SSOT)** repository pattern (`networkBoundResource`) reused by Ch 9 search, Ch 13 paging, Ch 29 offline-first, Ch 38 cart, and essentially every data-bearing feature.
>
> **Reference surface in StreamCart:** `:core:database` (Room) and the SSOT repositories in `:data:*` that back the catalog, cart, and watch history.

3.1 Problem Definition and Real-World Context

Where does the "truth" of your app's data live? In a naive app the answer is "whatever the last network call returned, held in a ViewModel field." That breaks the moment you have more than one screen, rotation, offline use, or background updates:

- **Two screens disagree.** The product list and the product detail each fetch independently; a price update shows on one and not the other. There is no single authority.
- **Flicker and loss on rotation / process death.** In-memory results vanish; the screen refetches and flashes a spinner.
- **No offline.** With the network as the source of truth, the app is useless on a train.
- **Write/read skew.** A write updates the server but not the local view, so the UI lies until the next refresh.

The fix is a **Single Source of Truth**: a local store (Room) that the UI observes, and a repository that is the *only* writer to it. The network becomes an *updater* of the local store, not the thing the UI reads. This is the `networkBoundResource` / offline-first read pattern, and it is the backbone of almost every feature in the book.

Three practical use cases

1. **E-commerce catalog (StreamCart Shop).** Product list and detail both observe Room; one fetch updates the DB; both screens update together and survive rotation and offline.
2. **OTT watch history (StreamCart Watch).** Playback progress is written locally first (instant UI), then synced; the "Continue Watching" row reads the DB and is always consistent across the app.
3. **Social feed cache (StreamCart Social).** The feed renders instantly from the cached DB on cold start, then refreshes from the network in the background.

Why this is architectural

SSOT answers *who owns state*, *who may write it*, *how the UI observes changes*, and *how reads and writes stay consistent* — the defining questions of the data layer. Get this right and features compose; get it wrong and every feature fights the same consistency bugs.

3.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	The UI observes data as a reactive stream (`Flow`) from the local store.
F2	The repository is the single writer to the store.
F3	Network responses update the local store, which emits to observers.
F4	Reads succeed offline from cached data.
F5	Writes are transactional (all-or-nothing).
F6	Schema changes are migrated safely without data loss.
F7	Expose load/refresh state (loading/error) alongside cached data.
F8	Support observing a single entity and lists.

Non-functional requirements

Attribute	Policy
Consistency	One authority (Room); no component bypasses the repository.
Performance	DB I/O off main thread; indexed queries; paging for large sets (Ch 13).
Reliability	Transactions; migrations tested; no partial writes.
Availability	Reads work offline.
Maintainability	DTO/entity/domain separation; mappers isolate layers.
Memory	Stream + page rather than load entire tables; `Flow` cancellation.
Security	Optional encryption at rest (Ch 24); no secrets in plaintext.

Scope

- **In scope:** Room setup, entities/DAOs, transactions, the SSOT repository pattern (`networkBoundResource`), `Flow` observation, migration basics.
- **Out of scope:** full offline sync engine (Ch 29–33), paging (Ch 13), DataStore for key-values (Ch 4), encryption specifics (Ch 24), FTS (Ch 35).

Assumptions & constraints

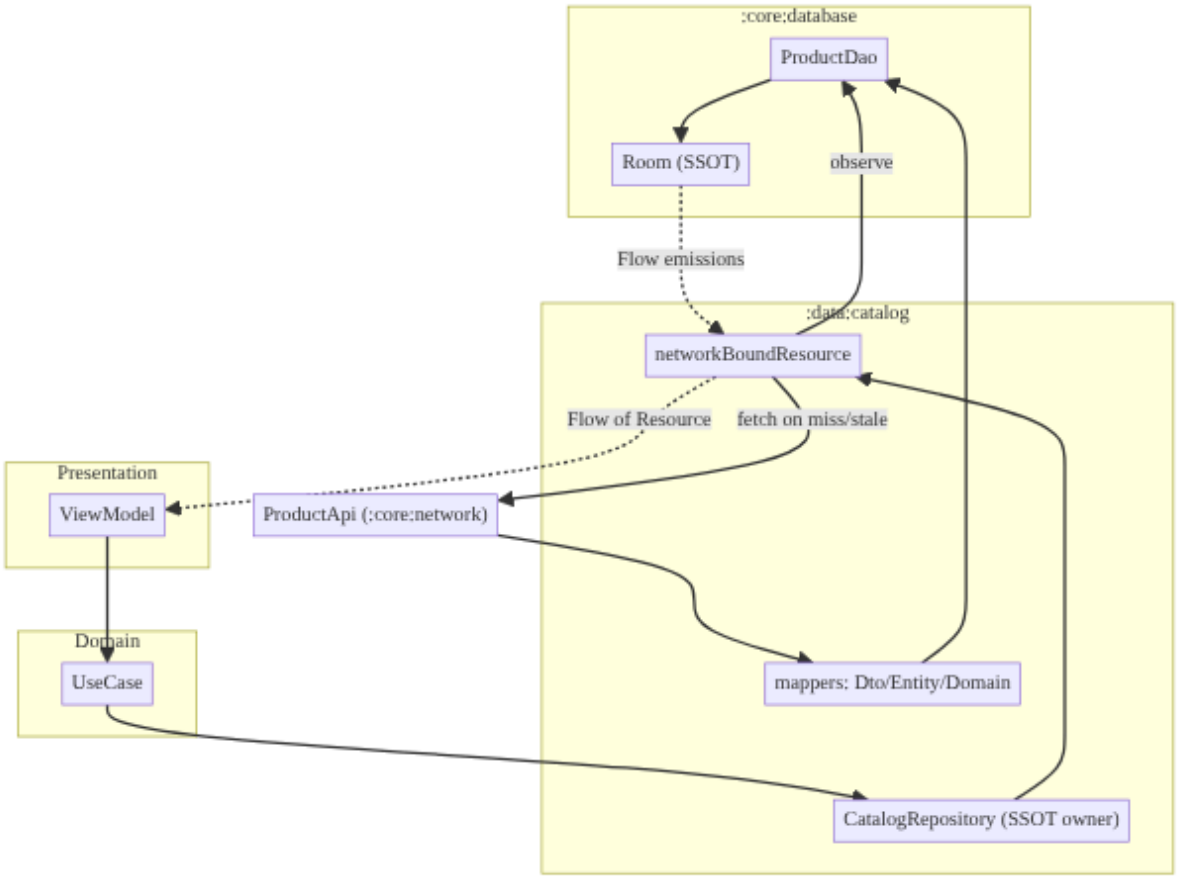
- Room 2.6+, Kotlin, coroutines/Flow, Hilt. Three model layers: `Dto` (network), `Entity` (Room), domain (`Product`). `minSdk 24`.

3.3 System Design Fundamentals

- **Single Source of Truth.** One component authoritatively holds each entity's state. The UI reads only from it. In StreamCart that's the Room-backed repository.
- **Reactive reads.** Room DAOs can return `Flow<T>`; Room re-emits whenever the queried tables change, so a write anywhere updates every observer — this is what keeps two screens consistent.
- **Unidirectional updates.** Network → repository → Room → Flow → UI. Writes go UI → repository → Room (→ network). The UI never writes the DB directly.
- **Transactions & atomicity.** Multi-step writes (e.g., replace a list) run in a single transaction so observers never see a half-updated state.
- **Mapping layers.** `Dto ↔ Entity ↔ Domain` keeps network schema, storage schema, and domain model independent — each can evolve without breaking the others (Ch 16).
- **`networkBoundResource`.** The canonical pattern: emit cached data, optionally fetch, persist the result, and let Room re-emit — yielding offline-first reads with fresh-when-possible data.

Core insight: **make the database the source of truth and the network a writer to it.** Observation (`Flow`) then makes consistency automatic instead of something each feature must coordinate.

3.4 High-Level Architecture

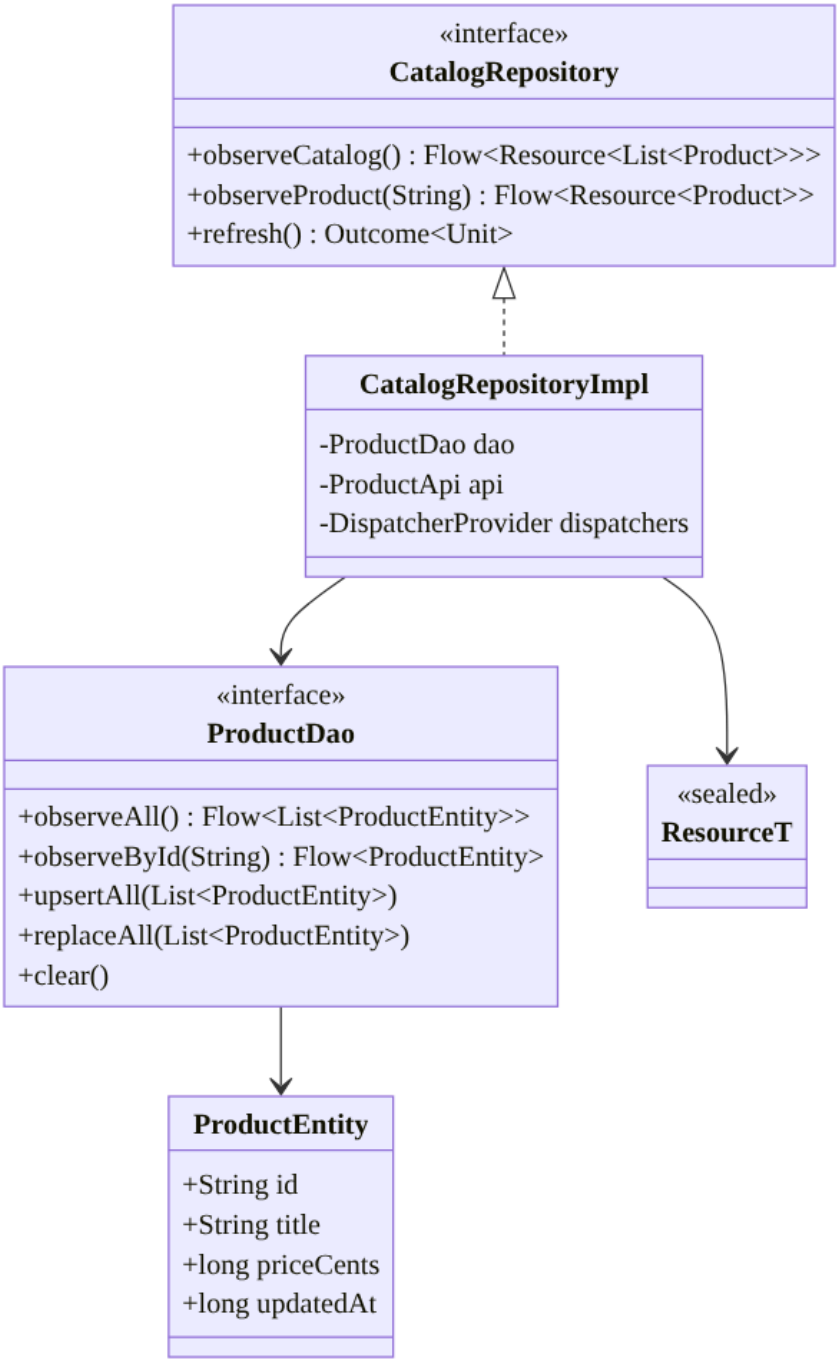


Components & ownership

- **Room (`:core:database`)** is the **SSOT**. `ProductDao` exposes `Flow<List<ProductEntity>>` reads and suspend writes.
- **`CatalogRepository`** is the only writer; it runs `networkBoundResource` to combine cache + network and maps across layers.
- **`networkBoundResource`** emits `Resource<T>` (loading/success/error wrapping the cached data), fetches when stale, persists, and relies on Room to re-emit.
- **Network** provides DTOs via `:core:network`; mappers convert to entities for storage and to domain models for the UI.

Dependency direction: `:data:catalog → :core:database` and `→ :core:network`. The database module knows nothing of features or the network.

3.5 Detailed Component Design



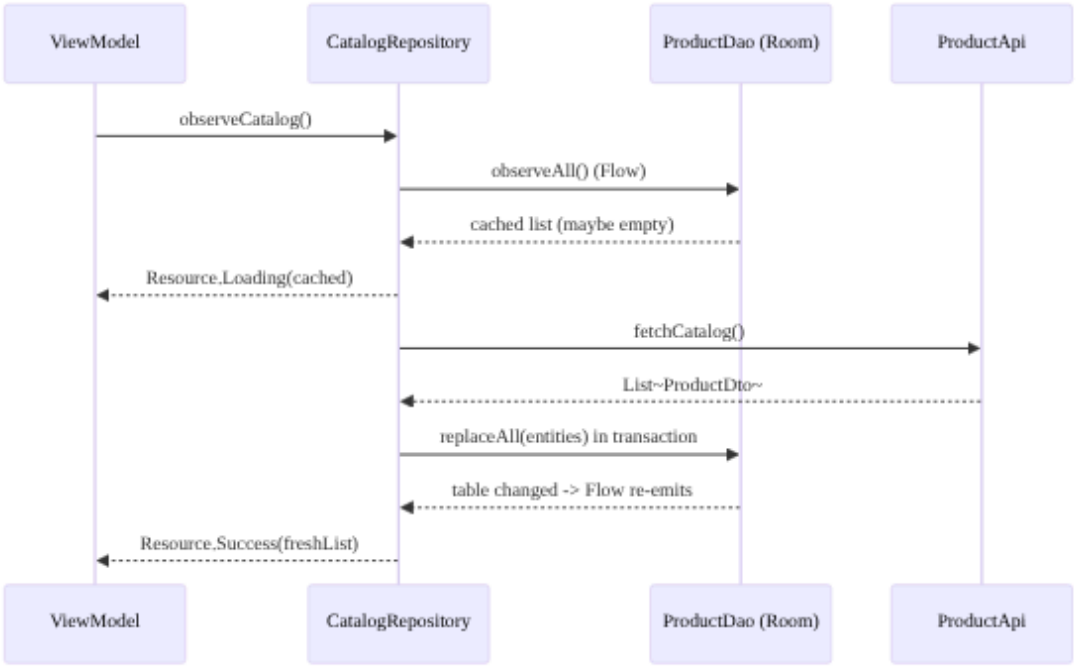
Component	Responsibility	I/O	State	Errors
`ProductEntity`	storage shape	—	—	—
`ProductDao`	typed DB access	queries/writes	Room-owned	SQLite exceptions
`CatalogRepositoryImpl`	SSOT writer, cache+network orchestration	domain in/out	none (DB holds state)	maps to `AppError`
`Resource<T>`	cached data + load state	—	—	carries `AppError`

```
sealed interface Resource<out T> {
    data class Loading<T>(val data: T?) : Resource<T>        // cached data while refreshing
    data class Success<T>(val data: T) : Resource<T>
    data class Error<T>(val error: AppError, val data: T?) : Resource<T>  // cached data + error
}
```

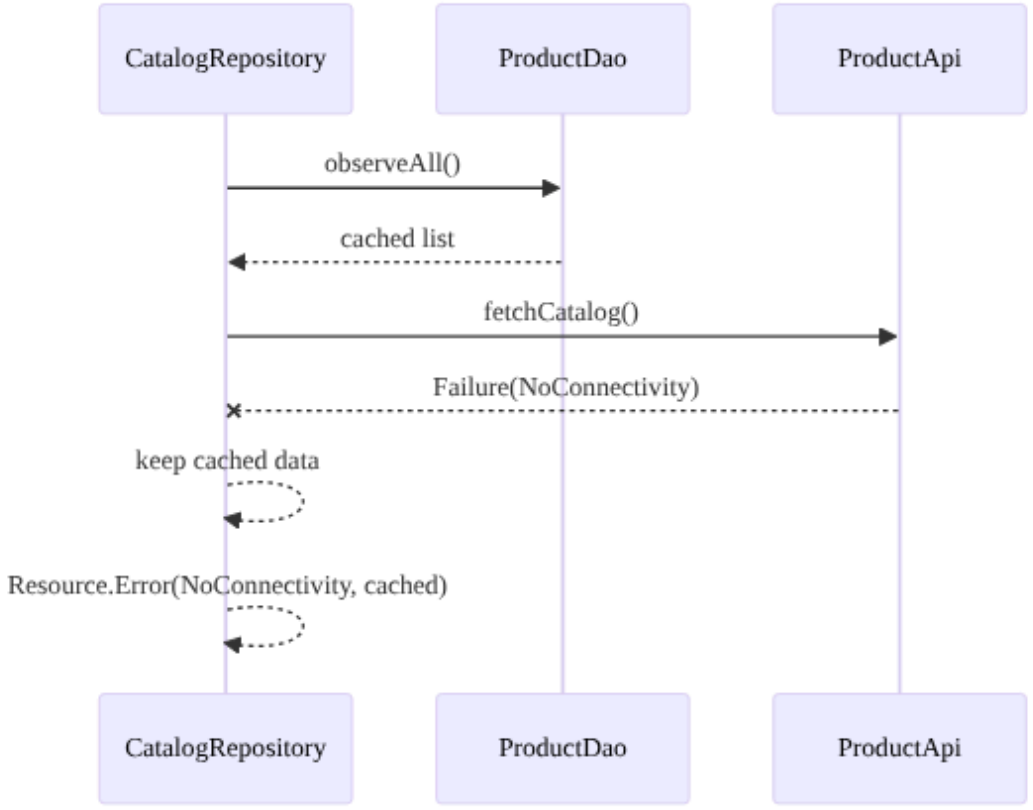
> `Resource` deliberately carries the cached `data` in *every* state so the UI can show stale-but-useful data during a refresh or after an error (the offline-first UX).

3.6 End-to-End Data Flow

Cold start with stale cache (offline-first read)



Network failure with cache present



3.7 Complete Kotlin Implementation

Package: `com.streamcart.core.database` and `com.streamcart.data.catalog`.

3.7.1 Entity & DAO

```
@Entity(
    tableName = "products",
    indices = [Index("title")],
)
data class ProductEntity(
    @PrimaryKey val id: String,
    val title: String,
    val priceCents: Long,
    val thumbnailUrl: String?,
    val updatedAt: Long,          // for staleness checks
)

@Dao
interface ProductDao {
    @Query("SELECT * FROM products ORDER BY title")
    fun observeAll(): Flow<List<ProductEntity>>

    @Query("SELECT * FROM products WHERE id = :id")
    fun observeById(id: String): Flow<ProductEntity?>

    @Upsert
    suspend fun upsertAll(items: List<ProductEntity>)

    @Query("DELETE FROM products")
    suspend fun clear()

    @Transaction
    suspend fun replaceAll(items: List<ProductEntity>) {
        clear()
        upsertAll(items)          // atomic: observers never see the empty interim
    }

    @Query("SELECT MAX(updatedAt) FROM products")
    suspend fun lastUpdatedAt(): Long?
}
```

> `@Transaction` on `replaceAll` is essential: without it, observers would momentarily see an empty table between `clear()` and `upsertAll()`.

3.7.2 Database & converters

```
@Database(entities = [ProductEntity::class, CartItemEntity::class], version = 1, exportSchema =
true)
abstract class StreamCartDatabase : RoomDatabase() {
    abstract fun productDao(): ProductDao
    abstract fun cartDao(): CartDao
}

@Module @InstallIn(SingletonComponent::class)
object DatabaseModule {
    @Provides @Singleton
    fun provideDatabase(@ApplicationContext context: Context): StreamCartDatabase =
        Room.databaseBuilder(context, StreamCartDatabase::class.java, "streamcart.db")
            .addMigrations(Migrations.M1_2)      // see 3.7.5
            .build()                             // no fallbackToDestructiveMigration in prod

    @Provides fun provideProductDao(db: StreamCartDatabase): ProductDao = db.productDao()
}
```

3.7.3 The `networkBoundResource` pattern

```
fun <Domain, Dto> networkBoundResource(
    query: () -> Flow<Domain>,                       // observe local SSOT
    fetch: suspend () -> Outcome<Dto>,               // network (via safeApiCall + reliability)
    saveFetchResult: suspend (Dto) -> Unit,          // persist into Room
    shouldFetch: (Domain) -> Boolean = { true },     // staleness/empty check
): Flow<Resource<Domain>> = flow {
    val current = query().first()
    if (shouldFetch(current)) {
        emit(Resource.Loading(current))
        when (val fetched = fetch()) {
            is Outcome.Success -> {
                saveFetchResult(fetched.data)        // triggers Room re-emission
                emitAll(query().map { Resource.Success(it) })
            }
            is Outcome.Failure -> {
                emitAll(query().map { Resource.Error(fetched.error, it) })
            }
        }
    } else {
        emitAll(query().map { Resource.Success(it) })
    }
}
```

3.7.4 Repository

```
interface CatalogRepository {
    fun observeCatalog(): Flow<Resource<List<Product>>>
    fun observeProduct(id: String): Flow<Resource<Product>>
    suspend fun refresh(): Outcome<Unit>
}

@Singleton
class CatalogRepositoryImpl @Inject constructor(
    private val dao: ProductDao,
    private val api: ProductApi,
    private val dispatchers: DispatcherProvider,
    private val clock: Clock,
) : CatalogRepository {

    override fun observeCatalog(): Flow<Resource<List<Product>>> =
        networkBoundResource(
            query = { dao.observeAll().map { list -> list.map { it.toDomain() } } },
            fetch = { safeApiCall(dispatchers.io) { api.fetchCatalog() } },
            saveFetchResult = { dtos ->
                dao.replaceAll(dtos.map { it.toEntity(clock.now()) })
            },
            shouldFetch = { cached -> cached.isEmpty() || isStale() },
        ).flowOn(dispatchers.io)

    override fun observeProduct(id: String): Flow<Resource<Product>> =
        dao.observeById(id)
            .map { entity -> entity?.let { Resource.Success(it.toDomain()) } ?:
Resource.Loading(null) }
            .flowOn(dispatchers.io)

    override suspend fun refresh(): Outcome<Unit> = withContext(dispatchers.io) {
        when (val r = safeApiCall(dispatchers.io) { api.fetchCatalog() }) {
            is Outcome.Success -> { dao.replaceAll(r.data.map { it.toEntity(clock.now()) });
Outcome.Success(Unit) }
            is Outcome.Failure -> Outcome.Failure(r.error)
        }
    }

    private suspend fun isStale(): Boolean {
        val last = dao.lastUpdatedAt() ?: return true
        return clock.now() - last > TTL_MS
    }
```

```
    private companion object { const val TTL_MS = 5 * 60_000L }
}
```

3.7.5 Migration

```
object Migrations {
    val M1_2 = object : Migration(1, 2) {
        override fun migrate(db: SupportSQLiteDatabase) {
            db.execSQL("ALTER TABLE products ADD COLUMN thumbnailUrl TEXT")
        }
    }
}
```

3.7.6 Mappers & module structure

```
fun ProductDto.toEntity(now: Long) = ProductEntity(id, title, priceCents, thumbnailUrl, now)
fun ProductEntity.toDomain() = Product(id, title, priceCents, thumbnailUrl)
```

```
:core:database
  StreamCartDatabase, Migrations, Converters
  dao/ ProductDao, CartDao
  entity/ ProductEntity, CartItemEntity
  di/ DatabaseModule
:data:catalog
  repository/ CatalogRepository(Impl)
  mapper/ ProductMappers.kt
  util/ networkBoundResource.kt, Resource.kt
```

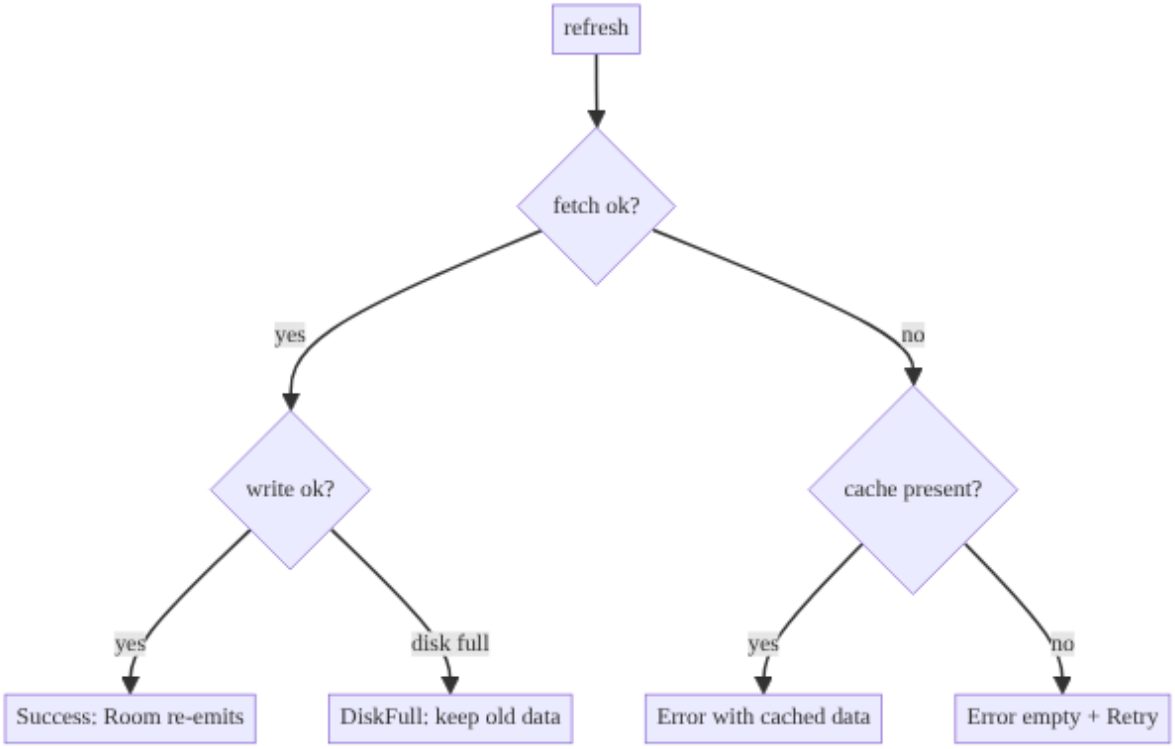
3.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Two screens observe same table	Divergent views	Both read the same DAO `Flow`; Room re-emits once on change
`clear()` then `upsert()` not atomic	Observers see empty list	`@Transaction replaceAll`
Concurrent writers (sync + user action)	Lost update / torn state	Room serializes writes; use transactions; SSOT = single writer repo
Fetch + local write interleave	Stale overwrite	Persist fetch result in a transaction; `updatedAt` for conflict (Ch 33)
Flow collected on main thread	Jank/ANR	`flowOn(io)`; Room suspend/Flow are main-safe but mapping should be off-main
Process death mid-write	Partial data	SQLite is ACID; a transaction either commits or rolls back

- **Room's write serialization** plus **transactions** gives correctness for concurrent writers; the SSOT rule (only the repository writes) prevents uncoordinated writes in the first place.
- **Reactive consistency:** because every reader observes the same DAO `Flow`, a single write propagates to all screens — no manual cross-screen notification.

3.9 Failure Scenarios and Recovery

Failure	Detection	Response	UX	Telemetry
Network down on refresh	`Outcome.Failure`	keep cache; `Resource.Error(data=cache)`	stale data + subtle error	`refresh_error{type}`
Empty cache + offline	cache empty + failure	`Error(data=null)`	offline empty state + Retry	`cold_offline`
DB disk full	`SQLiteFullException` → `DiskFull`	surface error; stop writes	"Storage full"	`db_error{type=disk_full}`
Corrupt DB	open fails	recreate (lose cache) or restore (Ch 37)	re-sync from network	`db_corrupt`
Migration failure	exception at open	fail safe (don't destroy in prod); report	forced resync path	`migration_fail{from,to}`
Partial write	transaction	rolled back by SQLite	consistent old state	`tx_rollback`
Stale overwrite (concurrent edit)	`updatedAt` mismatch	conflict policy (Ch 33)	—	`conflict`



Never use `fallbackToDestructiveMigration()` in production — it silently wipes user data on a missing migration. Provide migrations; fail loudly in debug.

3.10 Scaling the Implementation

- **Level 1.** One screen reads a DAO `Flow`; repository does a simple fetch + `replaceAll`.
- **Level 2.** Multiple screens share the SSOT via `networkBoundResource`; consistency is automatic.
- **Level 3 — production.** Staleness/TTL, transactions, migrations, `Resource` states for offline-first UX, indexed queries, encryption (Ch 24).
- **Level 4 — large.** Paging 3 + `RemoteMediator` for huge tables (Ch 13); FTS for search (Ch 35); partial updates via upsert; delta sync (Ch 32).
- **Level 5 — enterprise.** Full sync engine with outbox + conflict resolution (Ch 29–33), multi-module databases or a shared schema contract, schema governance and versioned migrations across app versions.

Avoid overengineering: not every entity needs `networkBoundResource` — purely remote, non-cached, one-shot reads (e.g., a legal document) can skip the DB.

3.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Source of truth	Room (DB)	In-memory cache	DB for persistence/offline/consistency; memory for ephemeral
Read pattern	`networkBoundResource`	fetch-then-return	NBR for offline-first; direct fetch for non-cached reads
Store	Room	DataStore (Ch 4)	Room for structured/relational/queryable; DataStore for key-values/prefs
Load state	`Resource<T>` carrying data	separate loading boolean	Resource keeps stale data visible; boolean loses it
Large lists	Paging 3 (Ch 13)	load all	Paging for big/remote sets; load-all for small bounded sets
Updates	`replaceAll`	diff/upsert by id	upsert to preserve unchanged rows and local edits; replaceAll for full snapshots
Encryption	SQLCipher (Ch 24)	plaintext	encrypt sensitive data (fintech/health); plaintext otherwise

`replaceAll` vs upsert-by-id: full replace is simplest for stateless lists but destroys local-only fields (e.g., a "liked" flag) and churns the table; upsert-by-id preserves rows and local edits and plays nicely with optimistic updates (Ch 72). Prefer upsert once you have local writes.

3.12 Security Considerations

- **Encryption at rest:** sensitive tables (payment tokens, PII) use SQLCipher or Jetpack Security (Ch 24); the key lives in Android Keystore (Ch 20).
- **`android:allowBackup`:** auto-backup can copy the DB to the cloud — exclude sensitive DBs via backup rules (Ch 37) so secrets don't leak off-device.
- **No secrets in plaintext columns**; tokens belong in encrypted storage, not Room.
- **Least data:** cache only what the UI needs; purge on logout (Ch 42).
- **Client vs backend:** the DB is a *cache/convenience*; the backend remains the authoritative record and must re-validate entitlements — a tampered local DB must never grant access.

3.13 Performance and Resource Optimization

- **Off main thread:** DAO suspend/Flow are main-safe for the query itself, but `flowOn(io)` moves mapping and collection work off-main; large mapping on the main thread causes jank.
- **Indexing:** add `@Index` on filtered/sorted columns; verify with `EXPLAIN QUERY PLAN` and the Database Inspector.
- **Paging:** never load huge tables into memory — Paging 3 (Ch 13).
- **Upsert over replace** to minimize write amplification and table churn.
- **Transactions** batch writes (one fsync) — far faster than N separate writes.

- **Profiling:** Android Studio **Database Inspector** (live queries), Room query logging, `PRAGMA` analysis, benchmark cold-start DB open time (Ch 88).

3.14 Testing Strategy

Use a **real in-memory Room DB** for DAO tests (catches SQL/transaction bugs); fake the API; test the repository with a real DAO + fake API.

```
class ProductDaoTest {
    private lateinit var db: StreamCartDatabase
    private lateinit var dao: ProductDao

    @Before fun setUp() {
        db = Room.inMemoryDatabaseBuilder(
            ApplicationProvider.getApplicationContext(), StreamCartDatabase::class.java
        ).allowMainThreadQueries().build()
        dao = db.productDao()
    }
    @After fun tearDown() = db.close()

    @Test fun `replaceAll is atomic - observers never see empty interim`() = runTest {
        dao.upsertAll(listOf(entity("1"), entity("2")))
        dao.observeAll().test {
            assertEquals(2, awaitItem().size)
            dao.replaceAll(listOf(entity("3")))
            assertEquals(listOf("3"), awaitItem().map { it.id })   // one emission, not empty-
then-one
            cancelAndIgnoreRemainingEvents()
        }
    }
}

class CatalogRepositoryTest {
    @Test fun `offline refresh keeps cached data and reports error`() = runTest {
        dao.upsertAll(listOf(entity("1")))
        val repo = CatalogRepositoryImpl(dao, FakeApi(fails = true), testDispatchers, FakeClock)
        repo.observeCatalog().test {
            val first = awaitItem()
            assertTrue(first is Resource.Loading && first.data?.size == 1)
            val err = awaitItem()
            assertTrue(err is Resource.Error && err.data?.size == 1)   // cache preserved
            cancelAndIgnoreRemainingEvents()
        }
    }
}

@Test fun migration_1_to_2() {
    helper.createDatabase(TEST_DB, 1).apply { close() }
    val db = helper.runMigrationsAndValidate(TEST_DB, 2, true, Migrations.M1_2)
    // assert new column exists
}
```

What to mock vs real: **real** Room (in-memory) + Turbine; **fake** API/clock. Always test migrations with `MigrationTestHelper` and `exportSchema = true`.

3.15 Production Observability

- **Metrics:** cache hit rate, refresh success/error rate, DB size, query latency (p95), migration success rate, disk-full error rate.
- **Logs:** `refresh{entity, result, correlation_id}`, `migration{from,to,ms}`, `db_error{type}`.
- **Tracing:** correlation ID from the network refresh links to backend; DB write spans for slow queries.
- **Debugging:** rising `migration_fail` after release → bad migration; growing DB size → missing purge/retention; cache hit rate drop → TTL too aggressive.
- **Privacy:** never log row contents (may be PII); log counts and types.

3.16 System Design Interview Preparation

> 40+ questions.

Fundamentals

1. *What is the SSOT and why?* The one authority for an entity's state (Room); keeps all screens consistent, enables offline, survives process death.
2. *Why make the DB the source of truth, not the network?* Offline reads, consistency across screens, no flicker, write/read coherence.
3. *How do two screens stay consistent?* Both observe the same DAO `Flow`; one write re-emits to all.
4. *What is `networkBoundResource`?* Emit cache → fetch if stale → persist → Room re-emits fresh; offline-first reads.
5. *Why `Resource<T>` carrying data in every state?* Show stale-but-useful data during refresh/errors.

Architecture 6. *Who may write the DB?* Only the repository (single writer). 7. *Why three model layers (DTO/entity/domain)?* Decouple network, storage, and domain so each evolves independently (Ch 16). 8. *Where does the DB module sit?* `:core:database`, depended on by `:data:*`; knows nothing of features/network. 9. *Direct fetch without caching — ever?* Yes, for non-cached one-shot reads; not every call needs the DB.

Kotlin/Android/Room 10. *How does Room make reads reactive?* DAO `Flow` re-emits on table changes. 11. *Why `@Transaction` on replace?* Atomicity — observers never see the empty interim. 12. *Are Room suspend/Flow main-safe?* The query runs off-main; use `flowOn(io)` for your mapping/collection. 13. `@Upsert` *vs* `@Insert(onConflict=REPLACE)`*?* Upsert = insert-or-update without deleting the row (preserves triggers/FKs more cleanly).

Concurrency 14. *Concurrent writers?* Room serializes writes; transactions keep multi-step writes atomic; SSOT limits writers to the repo. 15. *How avoid lost updates on concurrent edits?* `updatedAt`/version + conflict policy (Ch 33). 16. *Process death mid-write?* SQLite is ACID — commit or rollback.

Persistence 17. *How do migrations work?* `Migration(from,to)` SQL; test with `MigrationTestHelper`; `exportSchema`. 18. *Why never* `fallbackToDestructiveMigration` *in prod?* It wipes user data. 19. *Staleness strategy?* TTL via `updatedAt`, or ETag/`If-Modified-Since`, or server push invalidation. 20. *Encrypt the DB?* SQLCipher/Jetpack Security + Keystore for sensitive data (Ch 24).

Networking integration 21. *How does refresh update the UI?* Persist to Room → Flow re-emits; the UI never reads the network directly. 22. *Handle offline?* Serve cache; `Resource.Error(data=cache)`.

Scalability 23. *Huge tables?* Paging 3 + `RemoteMediator` (Ch 13); never load all. 24. *Search in the DB?* FTS (Ch 35). 25. *10M users — client DB concerns?* Size/retention, migration safety across versions, paging; backend scales separately.

Security 26. *allowBackup risk?* Sensitive DB can be backed up to cloud; exclude via backup rules (Ch 37). 27. *Can a tampered local DB grant access?* No — backend re-validates; the DB is a cache.

Performance 28. *Keep DB work off main?* `flowOn(io)`; index queries; page large sets. 29. *Reduce write cost?* Transactions (batch), upsert over replace. 30. *How profile queries?* Database Inspector, `EXPLAIN QUERY PLAN`.

Testing 31. *How test DAOs?* In-memory Room (real SQL). 32. *How test migrations?* `MigrationTestHelper.runMigrationsAndValidate`. 33. *What do you fake?* API/clock; keep Room real.

Production debugging 34. *Users report stale prices on one screen.* A component bypassed the SSOT or didn't observe the DAO `Flow`; enforce single-writer + shared observation. 35. *Crash spike after release — migration?* Missing/incorrect migration; add it, never destructive fallback. 36. *DB growing unbounded?* Add retention/purge; TTL eviction.

Trade-offs 37. *Room vs DataStore?* Room for structured/queryable data; DataStore for key-values/prefs (Ch 4). 38. `replaceAll` *vs upsert?* Upsert preserves local edits; replace for full snapshots. 39. *In-memory vs DB cache?* DB for persistence/offline; memory for ephemeral speed (can combine). 40. *When skip the DB entirely?* Non-cached, non-offline, one-shot reads.

Staff-level follow-ups

- "How do you keep a local-only `isLiked` flag through a server refresh?" (Upsert specific columns; don't `replaceAll`; or merge in the mapper — Ch 72.)
- "Observer sees two emissions on refresh (loading data, then success) — can you collapse flicker?" (`distinctUntilChanged`; keep data in Loading; UI shows data regardless of wrapper state.)
- "How do you evolve a column type safely across 3 app versions?" (Add new column, backfill in migration, dual-read, drop later — Ch 16/36.)

Design-change challenges

1. **Add offline writes.** Introduce an outbox (Ch 31) + optimistic DB write (Ch 72).
2. **Catalog grows to 1M items.** Paging 3 + `RemoteMediator` (Ch 13).
3. **Two devices edit the same entity.** Versioning + conflict resolution (Ch 33).
4. **Encrypt payment data at rest.** SQLCipher + Keystore (Ch 24); exclude from backup.
5. **Backend changes product schema.** Evolve DTO + migration; dual-read (Ch 16).

Mock walkthrough (abridged)

Clarify (offline? multi-screen? data size) → declare SSOT = Room → draw `networkBoundResource` → deep-dive transactions + reactive reads → failures (offline/disk/migration) → scale to paging/sync → defend Room vs DataStore and replace vs upsert.

3.17 Practical Exercises

1. **Beginner.** Back a product list with Room; prove two observers of the same DAO `Flow` update together on one write. *Eval:* single write, both update.
2. **Intermediate.** Implement `networkBoundResource` with TTL staleness; test offline keeps cache + reports error. *Eval:* `Resource.Error(data=cache)`.
3. **Advanced.** Switch `replaceAll` → upsert preserving a local `isLiked` column through refresh. *Eval:* flag survives server refresh.
4. **Production debugging.** Fix a repo whose `replaceAll` isn't transactional (observers flash empty). *Eval:* single non-empty emission.
5. **System design challenge.** Add a migration changing a column and test it with `MigrationTestHelper`; ensure no destructive fallback. *Eval:* migration validated; data preserved.

(Solutions withheld unless requested.)

3.18 Chapter Summary & Reference Sheet

Key decisions: Room is the **SSOT**; the repository is the **single writer**; the network **updates** the DB and Room re-emits; reads are **reactive** `Flow`**s**; multi-step writes are **transactional**; `Resource<T>` carries cached data in every state for offline-first UX; migrations are mandatory, destructive fallback forbidden.

Concepts: SSOT, reactive reads, `networkBoundResource`, transactions/atomicity, DTO/entity/domain mapping, staleness/TTL, upsert vs replace.

Common mistakes: network as source of truth; non-transactional multi-step writes; `fallbackToDestructiveMigration` in prod; mapping on the main thread; `replaceAll` destroying local edits; bypassing the repository to write the DB.

Checklist: ☐ DB = SSOT ☐ single writer ☐ DAO `Flow` reads ☐ `@Transaction` on multi-step writes ☐ `networkBoundResource` ☐ `Resource` carries data ☐ TTL/ staleness ☐ migrations + `MigrationTestHelper` ☐ `flowOn(io)` ☐ encrypt sensitive data.

Key APIs: `@Entity`, `@Dao`, `@Query`, `@Upsert`, `@Transaction`, `Flow`, `Room.databaseBuilder`, `Migration`, `MigrationTestHelper`, `inMemoryDatabaseBuilder`, `flowOn`, Database Inspector.

References:

- Room: https://developer.android.com/training/data-storage/room
- Room + Flow: https://developer.android.com/training/data-storage/room/async-queries

- Migrations: https://developer.android.com/training/data-storage/room/migrating-db-versions
- Guide to app architecture (data layer/SSOT): https://developer.android.com/topic/architecture/data-layer
- Offline-first: https://developer.android.com/topic/architecture/data-layer/offline-first

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/failure diagrams; full Room + DAO + `networkBoundResource` + repository + migration; concurrency; failure matrix; 5 levels; trade-offs; security (encryption/backup); perf; DAO + repo + migration tests; observability; 40 Q&A + follow-ups + 5 challenges. ✅
- **Expand later:** paging/`RemoteMediator` (Ch 13), full offline sync (Ch 29–33), encryption specifics (Ch 24), FTS (Ch 35).
- **Shared artifacts introduced:** `Resource<T>`, `networkBoundResource`, SSOT repository conventions — reused book-wide.